

HSMA: Holographic Spin-Manifold Architecture

A Layer-1 Protocol for Verifiable AI Consensus, Zero-MEV Execution, and Recursive State Folding

Technical Whitepaper v1.0 — Final Audited Specification — Errata Sync v1.0.1, Impl Sync v1.0.5

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Abstract

This document presents the formal specification, mathematical proofs, and implementation verification for the HSMA (Holographic Spin-Manifold Architecture) Layer-1 protocol. HSMA introduces a novel cryptoeconomic primitive: *Verifiable AI Inference as Proof of Useful Work (PoUW)*. By binding consensus voting weight to verified GKR matrix multiplications over the Pallas/Vesta Pasta curve cycle, the protocol converts wasted electrical hash-power into useful compute for enterprise AI clients.

HSMA eliminates Maximum Extractable Value (MEV) at the cryptographic layer via a threshold-encrypted mempool utilizing a *Commit-then-Simulate* manifest ceremony, where a 224-node committee (192 active + 32 standby, threshold $t = 112$) locks a beacon-shuffled ciphertext ordering *before* any decryption share is constructible. State transition integrity and historical data bloat are resolved using a HyperNova/Customizable Constraint Systems (CCS) folding architecture over the Pallas/Vesta 2-cycle, producing a succinct recursive epoch proof (size independent of transaction count) π_E verifiable by light clients in milliseconds.

The protocol's safety rests on a *combinatorial impossibility doctrine*: under adversarial weight $f < 0.20$, authenticated quorum forgery is structurally impossible ($\epsilon = 0$), not merely improbable. The total system soundness budget is $\epsilon_{\text{sys}} \approx 2^{-110}$ per epoch, dominated exclusively by the threshold-BLS unforgeability assumption. The C++20 reference implementation passes all conformance tests with **GATE GREEN**, the BLS12-377 group law is mathematically proven on 3 independent points, the BLS verification law is pairing-proven secret-free, and the M2 order-bound core is golden-proven end to end.

This document details: (1) the Pasta field arithmetic with wraparound soundness lemmas, (2) the Poseidon-3 hash and Sparse Merkle Tree state vault, (3) the Metastable Sub-Sampled Consensus with beacon-gated tie-breaking, (4) the Capped Cluster Hybrid weight engine, (5) the threshold-encrypted zero-MEV mempool, (6) the GKR/Sum-Check PoUW with Degree-2 LogUp activation tables, (7) the HyperNova NIVC state folding circuit, (8) the P2P gossip transport with beacon-seeded eclipse defense, (9) the end-to-end ϵ -budget composition theorem, and (10) the verified C++20 implementation with 25/25 conformance tests passing.

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1 Introduction

1.1 The Crisis in Decentralized Compute

Contemporary Layer-1 blockchains face an existential trilemma: they must balance decentralization, security, and scalability. Proof-of-Work (PoW) networks waste gigawatts of energy on meaningless SHA-256 hashes. Proof-of-Stake (PoS) networks centralize into oligopolies controlled by capital-rich entities. Furthermore, both architectures suffer from Maximum Extractable Value (MEV), wherein privileged sequencers front-run user transactions, extracting rent without providing value.

Simultaneously, the global demand for Artificial Intelligence (AI) compute has skyrocketed. Enterprises require verifiable, distributed matrix multiplication (GEMM) for model inference, but rely on centralized Web2 API providers, introducing single points of failure and censorship risks.

1.2 The HSMA Solution

HSMA resolves these crises by integrating three breakthrough cryptographic primitives:

1. **PoUW Consensus:** Validator weight is derived from verified GKR/Sum-Check proofs of AI inference, secured by a Capped Cluster Hybrid weight function that combines compute throughput Φ and staked capital S .
2. **Zero-MEV Encrypted Mempool:** A 224-member ($t=112$) threshold committee encrypts transactions and commits to their ordering *before* decrypting them, mathematically eliminating front-running.
3. **Holographic State Folding:** HyperNova IVC folds heterogeneous state transitions into a constant-size proof verifiable by light clients in milliseconds.

1.3 The Three Impossibility Principles

HSMA's design rests on three structural invariants:

Definition 1 (Order-Before-Knowledge (INV-M2-1)). *No party shall be able to learn transaction payload contents before that transaction's execution position is irreversibly committed, and no party shall be able to alter an execution sequence after learning contents. Enforcement: payload encryption against a threshold committee key whose decryption shares are bound to the committed ordering root (attribution and replay-freedom), with pre-freeze threshold-scale decryption prevented by counting: adversary identities $\leq 100 < t$ (DEC-057).*

Definition 2 (Combinatorial Quorum Impossibility). *Under adversarial weight fraction $f < 0.20$, authenticated sampling responses, frozen weight snapshots, and the enforced sampling floor $\varphi_{\text{floor}} = 0.50 > f/\alpha$ (with quorum threshold $\alpha = 0.75$), the observation of an adversarial supermajority in any consensus sample is **impossible**: $\epsilon_{\text{quorum}} = 0$ identically.*

Definition 3 (Succinct Holographic Accumulation). *Verification of full-ledger integrity shall require only: (i) the epoch header certificate chain, and (ii) one succinct epoch proof π_E of size independent of transaction count.*

The phrase “**impossible beats punishable**” names the design heuristic: wherever a draft specification proposed deterring misconduct (slashing a cheater), review demanded its promotion to unconstructibility (cheating has no valid witness).

2.5 Poseidon-3 Hash

All SNARK-friendly hashing uses Poseidon over $\mathbb{F}_p$ with $t = 3$ (rate 2, capacity 1), $\alpha = 5$, $R_F = 8$, $R_P \geq 56$. The MDS matrix is a Cauchy construction with machine-checked non-singularity of all minors. Round constants are derived via SHA-256 counter DRBG with rejection sampling. Domain separation is enforced via per-tree Initialization Vectors:

Role	Tag / IV
State tree nodes	IV_STATE_NODE
State tree leaves	IV_STATE_LEAF
Order tree	IV_ORDER
Decree tree	IV_DEGREE
Weight tree	IV_WEIGHT
Contact root	IV_CONTACT
Revocation root	IV_REVOCATION
Vote preimage	HSM_MSSC_VOTE_v1
Ordering lock	HSM_ORDER_v1
Decryption share	HSM_DEC_SHARE_v2
Epoch beacon	HSM_BEACON_V1
Hash-to-G1	HSM_H2G1_V1
M2 key derivation	HSM_KDF_V1
M2 DEM keystream/tag	HSM_DEM_V1 / HSM_DEM_TAG_V1
M2 ciphertext hash	HSM_CT_V1
M2 order root	HSM_ORDROOT_V1
M2 user authorization	HSM_SIG_USER_V1
Sum-check transcript	HSM_SUMCHECK_v1
CycleFold hash (Vesta)	HSM_CYCLEFOLD_v1
Vesta state tree node	IV_STATE_NODE_V
Vesta state tree leaf	IV_STATE_LEAF_V
Vesta decree tree	IV_DEGREE_V

2.6 Sparse Merkle State Vault

Account state is maintained in a Sparse Merkle Tree (SMT) of depth $D_S = 64$.

Leaf schema: $(m_{\text{bal}}[126] || s_{\text{bal}}[1] || \text{nonce}[64] || \text{flags}[1]) = 192 \text{ bits} < 2^{254}$, packed into a single field element, hashed under IV_STATE_LEAF; keyed $K_a = \text{Poseidon}(\text{"HSM_ACCT"} || \text{sender_pk})$.

Empty-subtree precomputation: $E_0 = \text{Poseidon}(\text{IV_STATE_LEAF}, 0, 0)$; $E_{d+1} = \text{Poseidon}(\text{IV_STATE_NODE}, E_d, 0)$. The constants $\{E_d\}_{d \leq 64}$ are compile-time circuit constants: untouched branches cost zero constraints.

Two-tier elision:

1. Empty-wedge propagation: $\text{empty} \oplus \text{empty}$ ascends as Empty with zero allocations.
2. Unchanged-subtree short-circuit: ascent halts when a recomputed hash equals the old spine hash (root-identity theorem).

Leaf width is the exact 24-byte window of DEC-126 (Errata E-11). **Leaf binding (CA-72 closure):** $L(K, v) = \text{Poseidon}(\text{IV_STATE_LEAF}, K, v)$, binding the key to the payload to prevent index-collision malleability.

3 Metastable Sub-Sampled Consensus (MSSC)

3.1 Protocol Overview

HSMA utilizes an Avalanche-family consensus protocol. Nodes query $k = 20$ peers (escalating to $k = 40$ after 1,000 rounds) weighted by W_i . If the quorum threshold $\alpha = 0.75$ is met for $\beta = 150$ consecutive rounds, the transaction is finalized. A floor guard $\varphi_{\text{floor}} = 0.50$ aborts rounds where sampled weight $< 50\%$ of expected.

3.2 Canonical Vote Preimage (DEC-016)

MSSC responses are signed under the exact domain-separated preimage:

$$\sigma = \text{Sign}_{\text{HSM_MSSC_VOTE_V1}}(\text{epoch}_E \parallel \text{epoch_weight_root}_E \parallel C_{id} \parallel r \parallel \text{preference}) \quad (2)$$

No timestamps, no view state, nothing prover-discretionary. Two honest responses for identical (epoch, root, C , r , pref) are byte-identical and signature-identical.

3.3 Safety: Authenticated Quorum Impossibility

Lemma 7 (Quorum Impossibility). *Under snapshot freshness (H_4) and canonical preimages: for any sample, adversarial mass $\leq f \cdot W_{\text{total}}$ while sampled mass $\geq \varphi_{\text{floor}} \cdot W_{\text{total}}$; hence sampled adversarial fraction $\leq f/\varphi_{\text{floor}} = 0.40 < \alpha = 0.75$. An adversarial quorum observation is impossible.*

Proof. Countability forces $W_{\mathcal{A}}^{\text{tallied}} \leq W(\mathcal{A}) \leq f \cdot T_E$: a vote contributes exactly its snapshot weight, and forged identities contribute zero because they open no Merkle path. The floor forces $S \geq \varphi_{\text{floor}} \cdot T_E$. Division yields $W_{\mathcal{A}}^{\text{tallied}}/S \leq f/\varphi_{\text{floor}} = 0.40 < 0.75 = \alpha$. $\square$

The inequality $\varphi_{\text{floor}} > f/\alpha$ (i.e., $0.50 > 0.267$) is a circuit-checked, ledger-enforced safety invariant (DEC-038).

3.4 Beacon-Gated Tie-Breaking

If a conflict set stalls for 50 rounds, it suspends until the next epoch's threshold-BLS beacon ($E + 1$) is revealed, then terminates deterministically:

$$\text{winner} = \arg \max_{tx \in C} H(\text{beacon}_{E+1} \parallel tx_{id}) \quad (3)$$

Because the beacon does not exist when the conflict is created, offline grinding of transaction identities against the breaker is **impossible**. Membership freezes at suspension (DEC-051); ties resolved by lexicographic tx_{id} (DEC-052).

Implementation (Step 11, DEC-200): `sim_beacon` is retired: the consensus goldens consume the real `HSM_BEACON_V1` threshold chain over the DKG committee — the first cross-pillar golden linkage (one committee testifies across DKG, threshold signing, the pairing, and the consensus sampler seed). The stagger delay is now computed by real threshold cryptography ($22 \rightarrow 376$ in the golden trace).

3.5 Liveness Analysis

Per-round metastable drift: $e^{-k \cdot D(\alpha \parallel 1/2)}$ where $D(0.75 \parallel 0.5) = 0.1887$ bits/round. For $k = 20$: drift $= e^{-20 \times 0.1308} = 0.0730$.

Expected rounds to $\beta = 150$ consecutive confirmations at persistence q :

$$E[T] = \frac{1 - q^\beta}{(1 - q) \cdot q^\beta} \quad (4)$$

At $q = 1 - 10^{-3}$ (honest unanimity): $E[T] = 162$ rounds (sub-second at ms ticks). Critical persistence $q^* \approx 0.975$: above, β -counter finalizes; below, stall breaker governs. Worst-case resolution is epoch-cadence-bounded.

4 Cryptoeconomics & Validator Admission (S1)

4.1 The Capped Cluster Hybrid Weight Function

Validator consensus weight W_i is a Capped Cobb-Douglas Hybrid:

$$\tilde{W}_c = \left\lfloor (\Phi_c^6 \cdot S_c^4)^{1/10} \right\rfloor \quad (5)$$

Where Φ_c is the verified AI inference FLOP throughput for cluster c , and S_c is the effective capital stake. The exponents ($\alpha = 0.6$ for compute, $1 - \alpha = 0.4$ for capital) are rigorously bounded by the width invariant $6a + 4b + \epsilon_{\text{guard}} \leq 126$ (DEC-019).

4.2 Sybil Resistance & Single-Pass Cap

Because the Cobb-Douglas function is homogeneous of degree 1, splitting into k identities preserves aggregate weight exactly. HSMA therefore enforces:

1. **Cluster-level capping (DEC-013):** Entities linked by payout-graph heuristics and signed independence attestations are capped as one cluster: $W_c = \min(\tilde{W}_c, W_{\text{max}})$.
2. **Single-pass cap (DEC-012):** $T_0 = \sum \tilde{W}_c$ (uncapped total, one pass); $W_{\text{max}} = 0.15 \cdot T_0$; $W_c = \min(\tilde{W}_c, W_{\text{max}})$.
3. **Dust pruning:** Clusters below $w_{\text{floor}} = 0.002 \cdot T_0$ are excluded from the active set and the denominator.
4. **Intra-cluster distribution (DEC-020):** Capped cluster weight distributed via largest-remainder rounding, ties by ascending `validator_id`.

4.3 Self-Dealing Exclusion (DEC-014)

Φ_i accrues only for workloads where:

1. Fees are non-refundable and completely burned.
2. Fee $\geq \kappa_{\text{floor}}$ (protocol minimum FLOP price).
3. The paying account is strictly outside the validator's declared cluster.

4.4 Slashing & Exit Queue

Violation	Evidence	Penalty
MSSC Equivocation	Two conflicting canonical signatures	100% burn + expulsion
False PoUW	GKR proof fails on-chain verifier	$\max(10 \times \text{fee}, \kappa \cdot B_{\text{min}})$
Manifest Equivocation	Two conflicting committee-signed manifests	50% burn across signing keys
Liveness Inactivity	≥ 2 corroborated missed queries	Weight decay $\rightarrow$ bond slash

Exit queue: 21-day unbonding delay; weight drops to 0 at $E + 1$; capital escrow extends 14 days past unbonding (E_{ev}) for late-arriving equivocation evidence. Light clients sync from socially pinned IVC checkpoint proofs, deprecating genesis sync.

5 Zero-MEV Encrypted Mempool (M2)

5.1 The Commit-then-Simulate Pipeline

Traditional *Simulate-then-Commit* sequencing requires the committee to read plaintext transactions before ordering them, creating a front-running cartel. HSMA inverts this to **Commit-then-Simulate** (DEC-025):

1. **Intake:** Envelopes $\{\text{sender_pk}, \text{nonce}, \text{fee_escrow}, \text{ct}\}$ are gossiped (G1 class P1).
2. **Lock:** $\text{order} = \text{Sort}(H(\text{"HSM_ORDER_v1"} \parallel \text{beacon}_E \parallel H(\text{ct})))$ — committee signs **order_root**.
3. **Share:** Order-bound decryption shares published (preimage: $\text{HSM_DEC_SHARE_v2} \parallel \text{epoch} \parallel \text{order_root} \parallel \text{ct_hash} \parallel \text{share}$).
4. **Aggregate:** Anyone reconstructs; $\text{decree_root} = \text{Merkle}(\text{ct_hash} \parallel \text{pt_hash} \parallel \text{status})$, certified by t -of- n threshold BLS.
5. **Simulate:** Deterministic re-simulation; insolvent entries $\rightarrow$ SKIP records.
6. **Execute & Fold:** **exec_root** committed; Pillar #4 consumes $(\text{order_root}, \text{decree_root}, \text{exec_root})$.

Implementation (Phase 0, Step 12, DEC-201..203): the G2-plane threshold KEM with the unified trust anchor $X_E = Y_2 = [S]G_{2,\text{gen}}$ (DEC-030): the sender picks r , publishes $R = [r]G_{2,\text{gen}}$, and derives $k = \text{SHA256}(\text{HSM_KDF_V1} \parallel \text{ser}([r]X_E) \parallel \text{ser}(X_E) \parallel \text{header})$ — from the shared secret, never the public R (the $k_2 = k$ end-to-end golden is the machine check of exactly this). Members release order-bound records: $D_j = [S_j]R$, verified by the pairing identity $e(G_1, D_j) = e(Y_{1,j}, R)$ against the Feldman G1 publics (item 6: closed), with attestations $\sigma_j = [S_j]H(\text{HSM_DEC_SHARE_V2} \parallel \text{epoch} \parallel \text{order_root} \parallel \text{ct_hash} \parallel \text{ser}(D_j))$ verified by $e(\sigma_j, G_{2,\text{gen}}) = e(H(\text{pre}), Y_{2,j})$. The aggregate $D = \sum_j \lambda_j D_j = [S]R$ re-derives k and opens the Phase-0 DEM (SHA-256-CTR keystream + tag; DEC-088's AES-GCM/ChaCha profiles deferred to the production step). All golden-proven on both sides of the language boundary: encrypt, order under the real beacon (first shuffle), verify, aggregate, decrypt — plus three negative cases (malformed share, wrong **order_root**, tampered ciphertext).

5.2 Threshold Parameters

$n = 224$ (192 active + 32 standby), $t = 112$. With $f = 0.20$: expected adversary seats $\mu = 44.8$, $\sigma \approx 6.0$.

- Confidentiality breaks at ≥ 112 colluders (50%): structurally impossible since eligible adversary identities cap at $m \leq f/w_{\text{floor}} = 100 < 112$ (DEC-039).
- Decryption halts at ≥ 113 unavailable seats (50.4%): adversary controls ≤ 100 ; halt requires ≥ 13 simultaneous honest failures — both extremes carry $\geq 12/13$ -*seatmargins*(*ErrataE-2*)*gins*.

5.3 Decryption Liveness & Refund Totality

If decryption fails: undecryptable envelopes SKIP with 100% escrow refund. User funds are never destroyed by committee faults (INV-M2-5). Three consecutive degraded epochs trigger emergency resharing.

5.4 Residual MEV Register

Vector	Status
Content front-running/sandwiching	Eliminated by INV-M2-1 + shuffle
Positional insertion	Eliminated by shuffle; grind-rate limited
Blind brackets on visible metadata	Residual; mitigable by stealth-address layer
Cross-domain latency arbitrage	Out of scope
Committee collusion $\geq t$	Structurally zero (Thm 4.6)

6 Verifiable AI Compute (PoUW & P5)

6.1 GKR/Sum-Check Protocol

Proof of Useful Work is scoped to Verifiable AI Inference (GEMM). The prover runs in $O(N \log N)$; the verifier runs in $O(\log N)$ (sub-millisecond). Per-proof soundness $\approx 2^{-128}$ with WHIR-class multilinear PCS at $\lambda = 128$ and mandatory per-epoch batched verification.

Implementation (Steps 19–20, DEC-212/213): the shared sum-check engine (dense MLEs, degree 1–2, chained-Poseidon transcript over `HSM_SUMCHECK_v1`) and the Phase-0 commitment layer are golden-proven; the GKR/GEMM wiring consumes them.

Neural network non-linearities (GELU, Sigmoid, Tanh) are evaluated via Degree-2 chunked LogUp-in-CCS lookup tables:

$$f(u) \approx T_0[u_{\text{high}}] + T_1[u_{\text{high}}] \cdot u_{\text{low}} + T_2[u_{\text{high}}] \cdot u_{\text{low}}^2 \quad (6)$$

Certified approximation: The 1-LSB ($\epsilon < 2^{-16}$) certification gate is strictly enforced by exhaustive grid enumeration during the table certification ceremony. Interpolation error drops to $O(h^3)$.

Rounding: Round-to-Nearest-Ties-to-Even (RNTE) is enforced via explicit witness constraints (q, r, b) :

$$b = [r > 2^{s-1}] \vee ([r = 2^{s-1}] \wedge (q \bmod 2)), \quad Y = q + b \quad (7)$$

6.2 Model Commitment Schema

$$\text{ModelCommit} = H(\text{dom} \parallel \text{arch_root} \parallel \text{weights_root} \parallel \text{quant_params} \parallel \text{table_root} \parallel \text{work_manifest} \parallel \text{fidelity_root} \parallel \text{version}) \quad (8)$$

6.3 Saturation Semantics (DEC-003)

Out-of-bounds accumulators clamp to $\pm(2^{31} - 1)$ and increment a verifiable saturation counter, aggregated into `fidelity_report_root`. Liveness is preserved; silent accuracy degradation is not — counters are public.

7 Holographic State Folding (P4)

7.1 Folding Stack

- Primary circuit: HyperNova over Pallas ($\mathbb{F}_p$), CCS
- Auxiliary: CycleFold over Vesta ($\mathbb{F}_q$) absorbs cross-curve commitment digests
- Non-uniform IVC: SuperNova-style NIVC over $\mathcal{F} = \{F_{\text{head}}, F_{\text{exec}}, F_{\text{close}}\}$
- Program counter: constrained variable with explicit transition matrix ($\text{head} \rightarrow \text{exec}^+ \rightarrow \text{close}$; all other edges unsatisfiable)

Implementation (Phase 0, Steps 14–21, DEC-207..214): the fold step family, the CCS constraint layer, the 41-byte NIVC accumulator, the epoch pipeline, the sum-check engine, the Phase-0 commitment layer, and the NIFS fold are golden-proven end to end; the production homomorphic PCS and HyperNova multifold fold *commitments* and are the remaining layer.

Per decree entry ℓ with certified tuple $(ct_hash_\ell, pt_hash_\ell, status_\ell)$:

- **Binding:** $a \cdot (\text{Poseidon}(\text{HSM_PT_v1}, pt_\ell) - pt_hash_\ell) = 0$
- **Nonce:** $a \cdot (\text{leaf_nonce}(\text{sender}_\ell) + 1 - pt.\text{nonce}) = 0$
- **Balance:** Sender debit, recipient credit via canonical registers with output range checks
- **Fee underflow protection (DEC-032):** Explicit LT gate: $\text{balance} \geq \Delta_{\text{req}}$ before subtraction
- **Self-send routing (DEC-031):** If $\text{sender} = \text{recipient}$, assert $\text{recipient_pre} == \text{sender_post}$
- **Padding-neutrality (DEC-033):** PAD rows bypass state verification entirely

7.2 Certificate Assumption (INV-P4-1)

The circuit assumes `decree_root` is valid; the verifier checks the threshold-BLS certificate out-of-circuit. Rationale: in-circuit pairing verification costs $\geq 10^6$ constraints; the verifier performs one aggregation check per epoch header anyway.

7.3 Proof-Carrying Data Boundary

Epoch proof π_E chains via `prev_digest` absorption in F_{head} . Light clients download epoch header ($height, state_root, \pi_E, \text{BLS_cert}$), verify the BLS signature (1 pairing), and verify π_E (HyperNova verifier, $O(1)$, ≈ 40 ms phone-class).

Proof size budget: $\pi_E \leq 73$ KB (WHIR wrap ≤ 72 KB + instances < 1 KB + digest < 0.1 KB), well within the 128 KB transport cap.

7.4 Async Fold Pipeline

Fold pipeline is asynchronous: $T_{\text{fold}} = 20\%$ of epoch. First certified π_E publisher earns δ_{fold} . Breach triggers alarms; the chain continues on full nodes — proof availability $\neq$ chain liveness.

8 P2P Layer & Network Integration (G1)

8.1 Beacon-Seeded Sampler Connectivity

Per-round rendezvous targets derived deterministically:

$$\sigma_r = H("HSM_PEERSEED_v1" \parallel \text{epoch_weight_root}_E \parallel \text{beacon}_E \parallel C_{id} \parallel r \parallel \text{AccountID}_u) \quad (9)$$

Queries dispatch over the direct link plus $R_{\text{rel}} = 2$ beacon-selected relay paths (distinct /24 and ASN). Response acceptance requires the canonical BLS preimage and an $O(\log N)$ weight inclusion proof against `epoch_weight_root`.

Containment (INV-G3): An adversary who fully eclipses u can only deliver data its weight already entitles it to emit. Eclipse $\Rightarrow$ stall/suspend, never divergent finalization.

Implementation (Step 18, DEC-211): the transport’s cryptographic layer is golden-proven: sampler determinism + weight proportionality (2000-draw histogram), taxonomy caps, the key-bound contact root with the 1-epoch rebind cooldown, and fixed-point EMA scoring (24h half-life, both directions); the socket adapter is the production transport step.

Class	Payload	Cap	Transport	Auth
P0 Safety	Slash evidence, certs	4 KB	Push-flood, zero delay	Sender-signed
P1 Consensus	MSSC votes	2 KB	gossipsub $D = 8$	Canonical preimage
P2 Manifests	ρ_E chunks	64 KB	Announce $\rightarrow$ pull	Content-addressed
P3 Bulk	π_E , GKR proofs	128 KB	Pull-only	Content-addressed

Informer reward (DEC-023): $\delta_{\text{info}} = 0.5\%$ of burn (cap 1000 tokens) to the first relay (60%) and the next seven relays within 30s (40%) of P0 slashing evidence (DEC-075 shared window) — converts censorship into an economically losing partition attack.

8.2 Contact Root & Anti-Churn

`contact_root` in epoch header; MSSC dials only pre-committed addresses. Rebinding cooldown $\Delta_{\text{rebind}} = 1$ epoch; reputation bound to `AccountID = Poseidon(consensus_pk)`, never to rotatable transport IDs.

9 End-to-End Security & ϵ -Budget

9.1 The A6 Composition Theorem

Under hypotheses H1–H5 (primitive security, honest preference consistency, closure window, snapshot freshness, circuit-spec equivalence):

$$\epsilon_{\text{sys}} = \underbrace{0}_{L1} + \underbrace{0}_{L2} + \underbrace{0}_{L4} + \underbrace{2^{-128}}_{L5} + \underbrace{2^{-127}}_{L6} + \underbrace{2^{-171}}_{\text{hash}} + \underbrace{2^{-110}}_{H1} \approx 2^{-110} \text{ per epoch} \quad (10)$$

Annualized (52,596 epochs/year at 600 s epochs):

$$\epsilon_{\text{year}} \leq 52,596 \cdot 2^{-110} \approx 2^{-94.3} \quad (11)$$

The 2^{-80} design target is cleared annually with 14 bits of margin.

9.2 Hypotheses

- H1** | BLS12-377 threshold-BLS EUF-CMA $\leq 2^{-110}$; co-CDH; Poseidon idealization; DDH+AEAD
- H2** | Honest preference consistency on common views
- H3** | Conflict propagation within $T_{\text{close}} < T_{\text{stall}}$ (empirical gate)
- H4** | Snapshot authenticity, staleness ≤ 1 epoch = 600 s
- H5** | Shipped circuits $\equiv$ these specifications (formal verification track)

9.3 Structural Zero Engineering

The zeros in the ϵ -budget are *engineered*, not inherent:

- **L1 (Quorum forgery):** $\epsilon = 0$ by authentication + frozen snapshots + $\varphi_{\text{floor}} > f/\alpha$ (DEC-038)
- **L2 (Committee takeover):** $\epsilon = 0$ by w_{floor} identity cap: $m_{\text{adv}} \leq 100 < t = 112$ (DEC-039)
- **L4 (Eclipse divergence):** $\epsilon = 0$ by INV-G3 containment

Remove the structural floor and the sortition term degrades to $\approx 2^{-73}$ (dust-spreading Poisson limit) — below target. The zeros are a parameter mandate, not a gift.

10 Implementation & Verification

10.1 C++20 Reference Implementation

The HSMA reference implementation is built in strict C++20 with zero floating-point operations in the numeric core (DEC-090). All cryptographic constants are mechanically derived from `scripts/gen_constants.py` (DEC-102); hand-transcription of curve constants is prohibited.

Repository structure:

```
1 hsma-core/
2 +-- include/hsma/          # Core protocol headers
3 | +-- fe.hpp               # Pallas base field (F_p)
4 | +-- poseidon.hpp         # SNARK-friendly hashing (Pallas domain)
5 | +-- fev.hpp              # Vesta base field (F_q)      GAP-03a (DEC-222)
6 | +-- poseidon_v.hpp       # Vesta-domain Poseidon-3    GAP-03b (DEC-223)
7 | +-- vault.hpp            # Sparse Merkle Tree state
8 | +-- consensus.hpp        # MSSC automaton
9 | +-- fold.hpp             # Fold step family (DEC-207)
10 | +-- ccs.hpp # CCS constraint layer (DEC-208)
11 | +-- nivc.hpp # NIVC fold accumulator, 41B bounded (DEC-209)
12 | +-- epoch.hpp # Epoch pipeline driver (DEC-210)
13 | +-- g1net.hpp # G1 transport skeleton (DEC-211)
14 | +-- sumcheck.hpp # Sum-check engine (DEC-212)
15 | +-- pcs.hpp # Phase-0 multilinear PCS (DEC-213)
16 | +-- nifs.hpp # NIFS fold (DEC-214)
17 | +-- lightclient.hpp # Light-client epoch-header certificate (DEC-215)
18 | +-- mempool.hpp         # Transaction gate
19 | +-- tx.hpp              # Transaction structure
20 | +-- params.hpp          # Protocol constants (static_assert enforced)
21 | +-- sha256.hpp          # FIPS-180 SHA-256
22 | +-- update.hpp          # SMT updates + proofs
23 | +-- threshold/         # Pillar-II threshold module (Steps 7-13)
24 | +-- scalar_r.hpp # BLS12-377 F_r Montgomery CIOS kernel
25 | +-- poly.hpp # DKG scalar semantics (Horner, Lagrange)
26 | +-- mont384.hpp # 384-bit Montgomery engine (promoted)
27 | +-- g1.hpp # G1 Jacobian arithmetic (promoted)
28 | +-- interop.hpp # Fr <-> fe6 interop
29 | +-- h2g1.hpp # Hash-to-G1 (DEC-189)
30 | +-- sig.hpp # Threshold BLS partials + aggregate (DEC-191)
31 | +-- beacon.hpp # HSM_BEACON_V1 epoch beacon (DEC-192)
32 | +-- dkg.hpp # Feldman DKG commitments (DEC-190)
33 | +-- fq2.hpp # F_q2 arithmetic, beta = 5 (DEC-193)
34 | +-- g2.hpp # E' twist curve over F_q2 (DEC-194..196)
35 | +-- fq12.hpp # F_q12 tower + tower-norm inverses (DEC-197)
36 | +-- pairing.hpp # twist + Miller + Tate + BLS verify (DEC-198/199)
37 | +-- m2.hpp # M2 order-bound decryption shares (DEC-201..203)
38 | +-- m2prod.hpp # sigma_user + per-member partials (DEC-204..206)
39 +-- conformance/         # Test suite (25/25 passing)
40 | +-- test_step2.cpp      # Field arithmetic tests
41 | +-- test_step3.cpp      # Numeric kernel tests
42 | +-- test_step4.cpp      # State vault tests
43 | +-- test_step5.cpp      # Mempool tests
44 | +-- test_step6.cpp      # Consensus tests
45 | +-- test_step7.cpp      # Threshold F_r kernel conformance
46 | +-- test_step8.cpp      # G1 promotion parity conformance
47 | +-- test_step9.cpp      # Threshold protocol layer (h2g1, DKG, beacon)
48 | +-- test_step10.cpp     # F_q2 + E' twist + G2 conformance
49 | +-- test_step10b.cpp    # Pairing layer: Miller, Tate, BLS law
50 | +-- test_step12.cpp     # M2 order-bound decryption shares
51 | +-- test_step13.cpp     # M2 production hardening
52 | +-- test_step14.cpp     # Fold step family conformance
53 | +-- test_step15.cpp     # CCS constraint layer conformance
54 | +-- test_step16.cpp     # NIVC fold accumulator conformance
```

```

55 | +-- test_step17.cpp # Epoch pipeline conformance
56 | +-- test_step18.cpp # G1 transport skeleton
57 | +-- test_step19.cpp # Sum-check engine
58 | +-- test_step20.cpp # Multilinear PCS
59 | +-- test_step21.cpp # NIFS fold
60 | +-- test_step22.cpp # Light-client certificate
61 | +-- test_step23.cpp # phi0-phi7 end-to-end epoch
62 | +-- test_step24.cpp # Vesta field twin (GAP-03a)
63 | +-- test_step25.cpp # Vesta-domain Poseidon (GAP-03b)
64 | +-- diag_step5.cpp      # Mempool diagnostics
65 | +-- diag_step6.cpp      # Consensus diagnostics
66 +-- libhsma_numcore/      # 127-bit scratchpad accumulator
67 | +-- scratch127.hpp
68 +-- src/
69 | +-- params_probe.cpp    # Identity probe (DEC-086 bot)
70 +-- tools/
71 | +-- bls_derive.cpp      # BLS12-377 curve generator
72 +-- scripts/
73 | +-- gate.sh            # Full diagnostic gate
74 | +-- gen_constants.py    # Montgomery constants generator
75 | +-- check_no_fp.sh      # Float-free lint (DEC-090)
76 | +-- add_smt.py          # SMT utilities
77 +-- CMakeLists.txt
78 +-- README.md
79 +-- docs/
80 | +-- whitepaper.tex      # This document
81 | +-- DECISIONS.md        # decision ledger: DEC-001..223 (Sections I-XXIX)

```

10.2 GATE GREEN Verification

The full diagnostic gate (./scripts/gate.sh) produces:

```

1 [lint] DEC-090 clean: numeric core is float-free.
2 [2/56] Building CXX object CM    all 32 generated headers present (each named +
   counted)
3 SEED: fault=0 allocs=66
4 LOOKUP: OK  bal=1000 nonce=0
5 TLEAF tag=2 (want 2)
6 REC   tag=3 (want 3)
7 STORED = 0x1c3cfad21077198c52c65724b158b408503a1e628d072c010f106554f4f23f9b
8 VERDICT: canonical-at-rest (DEC-123 held)
9 DIAG6: SAMPLE-DETERMINISTIC OK
10 DIAG6: CONTEXT-BINDING LIVE
11 DIAG6: QUORUM-PATH LIVE (kind=2)
12 DIAG6: BREAKER-DETERMINISTIC OK delay=98
13 DIAG6: VERDICT ALL-OK
14 10/25 Test #10: step10_conformance ..... Passed    0.08 sec
15 11/25 Test #11: step10b_conformance ..... Passed   14.60 sec
16 12/25 Test #12: step12_conformance ..... Passed   47.95 sec
17 13/25 Test #13: step13_conformance ..... Passed    7.43 sec
18 14/25 Test #14: step14_conformance ..... Passed    0.36 sec
19 15/25 Test #15: step15_conformance ..... Passed    0.17 sec
20 16/25 Test #16: step16_conformance ..... Passed    0.56 sec
21 17/25 Test #17: step17_conformance ..... Passed    0.17 sec
22 18/25 Test #18: step18_conformance ..... Passed    0.30 sec
23 19/25 Test #19: step19_conformance ..... Passed    0.07 sec
24 20/25 Test #20: step20_conformance ..... Passed    0.08 sec
25 21/25 Test #21: step21_conformance ..... Passed    0.05 sec
26 22/25 Test #22: step22_conformance ..... Passed    5.84 sec
27 23/25 Test #23: step23_conformance ..... Passed   23.31 sec
28 24/25 Test #24: step24_conformance ..... Passed    0.07 sec
29 25/25 Test #25: step25_conformance ..... Passed    0.04 sec
30 100% tests passed out of 25

```

```

31 Total Test time (real) = 102.09 sec
32 [emit] ./build/generated/consensus_golden.hpp (C1 fin@150, C2 sus@50, winner=H
    , delay=376)

```

Implementation sync v1.0.2 (2026-09-05): threshold module, Steps 7–8. The conformance suite now stands at **8/8** with **13 generated headers** (machine-checked per gate run). `step7_conformance` proves the BLS12-377 scalar-field F_r kernel (4×64-limb Montgomery CIOS, canonical range $[0, r)$): 64 DRBG pairs bit-exact against a Python big-integer oracle, edge cases, and a full DKG $n=5, t=3$ scalar trace (share derivation, Lagrange-at-zero, secret reconstruction, tamper divergence). `step8_conformance` proves the G1 promotion — the 384-bit Montgomery engine and Jacobian `Pdbl/Padd/Pmul` promoted verbatim from the derivation kernel with in-source laws preserved — via 48 F_q golden pairs, addition/doubling/scalar-multiplication triples against a Python *affine* EC oracle, 8 Tonelli roots, and algebraic self-proofs ($\text{Padd}(P, P) = \text{Pdbl}(P)$, $[a+b]P = [a]P + [b]P$, $[r]G = \infty$, $[\#E]P = \infty$). Constants provenance: r is verified against the kernel literals through the family identity $x^4 - x^2 + 1 = r$ with $x = 0x8508c00000000001$; $q = h_1 r + x$ carries MR-52 primality, widths 377/253, and low-limb ancestry; the generator is cofactor-cleared and $[r]G = \infty$ is machine-checked every gate run. New module: `include/hsma/threshold/` (`scalar_r.hpp`, `poly.hpp`, `mont384.hpp`, `g1.hpp`); `tools/bls_derive.cpp` remains the derivation authority (DEC-109) — the promoted module reproduces its outputs under golden parity. The verbatim gate log above records the v1.0 baseline (6/6); the suite and the gate have since grown together.

Implementation sync v1.0.3 (2026-09-09): Steps 9–12. The conformance suite now stands at **12/12** with **18 generated headers**. `step9_conformance` proves the threshold protocol layer (DEC-189..192): hash-to-G1 (SHA-256 counter preimage, even-y law, cofactor-clear, $[r]H = \infty$), Feldman DKG pairing-free (25/25 share equations), threshold signing with the $\sigma = [S]H$ harness law, and the `HSM_BEACON_V1` epoch beacon (53-byte preimage, chain ×4 golden). `step10_conformance` proves F_{q2} ($\beta=5$, generator-discovered), the twist $E' : y^2 = x^3 + (5, 1)$ over F_{q2} (search-discovered, $[r]G_2 = \infty$), and the G2 generator (DEC-193..196). `step10b_conformance` proves the pairing layer (DEC-197..199): the F_{q12} tower ($v^3 = \gamma = (5, 1)$, $w^2 = v$), the twist map, the Miller loop (final-vertical skip by denominator elimination, $r \mid q^6 + 1$), the reduced Tate pairing, and the BLS law $e(\sigma, G_{2,\text{gen}}) = e(H, Y)$ — **secret-free: the DEC-191 harness-secret law is RETIRED** — with tower-norm inverses ($\sim 2000\times$ vs. Fermat). Step 11 (DEC-200) retires `sim`: the consensus goldens consume the real `HSM_BEACON_V1` chain over the G7 committee — the first cross-pillar golden linkage (the beacon feeds the sampler seed; the stagger moved $22 \rightarrow 376$, the first consensus constant computed by real threshold cryptography). `step12_conformance` proves the M2 order-bound decryption core (DEC-201..203): the G2-plane threshold KEM (unified $X_E = Y_2$), the Phase-0 DEM, the real-beacon ordering lock (first shuffle), order-bound share records verified by the pairing identity and BLS attestations, the Lagrange aggregate, the decrypt roundtrip, and three negative cases.

Implementation sync v1.0.4 (2026-09-11): Steps 13–21. The conformance suite now stands at **23/23** with **29 generated headers**. Step 13 (DEC-204..206) hardens M2 to the production model: σ_{user} envelope authorization (pairing-verified, forgery rejected — item 5 closed), per-member partials + permissionless id-driven aggregation, and the explicit cross-epoch replay negative (item 4 closed). Steps 14–17 (DEC-207..210) build the Phase-0 folding stack: the fold step family $\{F_{\text{head}}, F_{\text{exec}}, F_{\text{close}}\}$ over the proven mempool semantics, the CCS constraint layer with the PC-matrix SAT-proof (item 7 closed), the 41-byte NIVC accumulator ($O(1)$ in entry count — the succinctness mechanism), and the epoch pipeline connecting the M2 ordering to the fold (`order_root` feeds the fold head). Step 18 (DEC-211) adds the G1 transport skeleton: the beacon-seeded sampler (deterministic, weight-proportional), the 4-tier taxonomy (the P3 128KB cap is `static_assert`-equal to the transport firewall), the contact root, and fixed-point EMA scoring. Steps 19–21 (DEC-212..214) build the proof machinery: the sum-check engine over dense multilinear extensions (chained-Poseidon Fiat–Shamir over

HSM_SUMCHECK_v1), the Phase-0 multilinear PCS (commit/open/verify with verifier-derivable points; the production dual-layer Pedersen+WHIR honestly deferred), and the NIFS fold — two verified computations compress into one, *exactly*, the operation that makes the 41-byte carrier an actual proof.

Implementation sync v1.0.5 (2026-09-15): Phase 1 opens, P1-01..04. The conformance suite now stands at **25/25** with **32 generated headers**. P1-01 (DEC-218) refactors the constants generator into `scripts/gen/` (per-step modules + `gen_common.py` shared library), byte-identical proven. P1-02 (DEC-219..221) verifies the interop bridge to canonical BLS12-377: G1 seed→r→q→curve→subgroup chain confirmed in pure arithmetic; the G2 twist divergence found (our $b' = (5, 1)$ vs. the reference $b_2 = (0, B)$, isomorphic but not identical) and the wire map ψ derived, verified against ark-bls12-377 0.6.0's own generator, and golden-pinned as `bridge_golden.hpp` — the external audit's interop warning answered with a function. P1-03 (DEC-222) adds `fev.hpp`, the Vesta base-field twin over $\mathbb{F}_q$ (CIOS Montgomery with overflow-safe u128 accumulation — CA-R112: the 255-bit modulus triggers an overflow that Pallas's 254-bit never does), with 516-case Vesta golden coverage closing the 0/2472 gap. P1-04 (DEC-223) adds `poseidon_v.hpp`, the Vesta-domain Poseidon-3 (same $T=3$, $R_F=8$, $R_P=56$, $\alpha=5$; constants via DRBG + rejection over q_{Vesta} ; Cauchy MDS all-minors-nonsingular). The Cycle-Fold foundation is **two-thirds complete**: field + Poseidon proven; curve operations and the absorption step (GAP-04) remain.

The `tools/bls_derive.cpp` tool mathematically proves the BLS12-377 curve group law:

- **Field arithmetic:** 2472 golden vectors passing (cross-validated against Python big-integer reference)
- **BLS12-377 order:** $\#E = h_1 \cdot r$ **PROVEN on 3 independent points** — requires ~ 380 consecutive correct point doublings per scalar multiplication
- **Canonical parameter:** $b = 1$ pinned (canonical BLS12-377)
- **Tonelli-Shanks:** Verified (8/8 bigint equivalence)
- **Exit code:** 0 ([done-bls])

Root Cause #7 (8C Doubling Bug): The dbl-2009-1 formula required $Y_3 = E \cdot (D - X_3) - 8C$. The v3 implementation computed $C_8 = \text{madd}(QC, \text{madd}(QC, C_c, C_c), C_c)$ producing $6C$ instead of $8C$. Every Y-coordinate from every doubling was off by $+2C$. Fixed: $C_2 = C + C$; $C_4 = C_2 + C_2$; $C_8 = C_4 + C_4$.

Root Cause #9 (rand_below Infinite Loop): The scalar field r is 253-bit, so $g_r[5] = 0$. The RNG filled all 6 limbs with random 64-bit values, making $v > g_r$ almost always. Fixed: clear high zero-limbs of the modulus before generating.

Root Cause #8 (Consensus-Fixture Stall): 30-iteration scalar-multiplication fixture loops exceeded the 90-second timeout; telemetry ruled out Tonelli-Shanks non-termination; fixture counts reduced ($30 \rightarrow 3$). Narrative: DECISIONS.md E-9.

Root Cause #10 (h2c Exhaustion): hash-to-curve candidate exhaustion during fixture generation; resolved in the derivation tool.

10.3 Conformance Suite

Implementation status (Phase 0). Items 1–8, and 12–13 are machine-validated today by the 25/25 conformance binaries — item 6 by Step 12's `verify_share`; item 4 (explicit cross-epoch replay negative) and item 5 (σ_{user} envelope authorization, forgery rejected) closed by Step 13; item 7 (SAT-proof: all illegal PC edges unsatisfiable) closed by Step 15's CCS constraint layer — together with the step 9–12 threshold modules and the Phase-0 folding stack (Steps 14–21:

fold semantics, constraints, the 41-byte accumulator, the epoch pipeline, the transport skeleton, the sum-check engine, the commitment layer, and the NIFS fold). Remaining production gates: the homomorphic Pedersen + WHIR dual-layer PCS (DEC-063/071) and the full HyperNova multifold (commitment folding), full CycleFold curve operations, graded-cell certification, light-client sync, and the HHI envelope guard — and activate together with those modules.

The release-blocking conformance suite validates:

1. RNTE ties $X = \pm 2^{s-1}$, both parities
2. Breaker hash-collision lexicographic path
3. Nonce-gap skips with exact 25/75 and 0/100 fee splits
4. Cross-epoch envelope replay rejected via header-bound DEM
5. Third-party σ_{user} fabrication rejected
6. Malformed partial decapsulation caught by pairing identity
7. SAT-proof that all illegal PC edges are unsatisfiable
8. Width-violation clamp-with-counter
9. Graded-cell certification bit-exact reproducibility
10. Light-client rejection of genesis-sync requests
11. Envelope-guard trip on injected HHI violation
12. F_r scalar-kernel golden parity vs Python big-integer oracle (step 7; implemented)
13. G1 promotion: affine-EC oracle triples + algebraic self-proofs (step 8; implemented)

11 Consolidated Parameters

Group	Parameters
Consensus	$k = 20 \rightarrow 40$ @1000r, $\alpha = 0.75$, $\beta = 150$, stall 50r $\rightarrow$ beacon breaker, $\varphi_{\text{floor}} = 0.50 > f/\alpha = 0$
Committee	$n = 224$ (192+32), $t = 112$, ≤ 16 seats/cluster, $w_{\text{floor}} = 0.002T \Rightarrow m_{\text{adv}} \leq 100 < t$
Weight	$\tilde{W} = \text{RNTE}((\Phi^6 S^4)^{1/10})$, width $6a + 4b \leq 126$, $W_{\text{max}} = 0.15T_0$ single-pass
Mempool	Envelope $\{pk, nonce, escrow\}$ clear; KEM/AEAD bound to $(epoch, sender, nonce)$; order = b
Circuits	Pasta+CycleFold; CCS/HyperNova multifold; PC matrix $\{head, exec^+, close\}$; SMT $D_S = 64$
Economics	$B_{\text{min}} = 50k$; burn fees $\geq \kappa$; $\delta_{\text{info}} = 0.5\%$ (cap 1000); δ_{fold} ; unbond 21d + 14d horizon; activation
Error budget	Structural zeros $(L1/L2/L4) \cdot 2^{-128} \cdot 2^{-127} \cdot 2^{-171} \cdot \text{assumption } 2^{-110} \Rightarrow \epsilon_{\text{sys}} \approx 2^{-110}/\text{epoch}$
Epoch	600 s nominal; checkpoints $K = 6$ (hourly); $T_{\text{fold}} = 120$ s

12 Limitations and Honest Disclosure

Metadata (sender, nonce, fee) remains visible for routing and pricing. Cross-domain MEV persists beyond any single-chain remedy. Threshold-BLS and KEM assumptions are quantum-vulnerable; post-quantum migration is reserved and unscheduled. The Holographic Boundary eliminates *verification* storage, not data-availability duties. Prover concentration for π_E assembly is mitigated by a permissionless publisher market, not eliminated. Liveness constants are simulation-calibrated (Phase-1 wind tunnel), not analytic. The edifice assumes H5 — shipped

circuits equal these specifications — whose formal-verification track is a release blocker. HSMA is Phase 0: a verified bare-metal core; the threshold module is complete through the pairing, the M2 order-bound core, and its production hardening (Steps 7–13); the folding arc is complete in Phase-0 form (Steps 14–21: semantics, constraints, the 41-byte accumulator, the pipeline, the sum-check engine, the commitment layer, the NIFS fold — suite 25/25); the production proof system (homomorphic PCS + HyperNova multifold), full CycleFold curve operations, and the remaining release gates are specified, not yet built.

13 Conclusion

HSMA’s contribution is not any single primitive but a composition discipline: every subsystem hands clean, certificate-authenticated artifacts to a single fold boundary, every security claim reduces either to a structural zero or to a named cryptographic assumption, and every residual weakness is enumerated rather than rhetorically buried.

The result is a ledger whose safety argument fits on one line — $\epsilon_{\text{sys}} \approx 2^{-110}$, assumption-dominated — and whose liveness recovers from every network-level attack by construction. Continuous-state throughput, MEV-immune ordering, and useful-work security stop being trade-offs when ordering is separated from knowledge, safety is separated from statistics, and history is separated from verification.

The C++20 reference implementation passes all 25 conformance tests with **GATE GREEN**. The BLS12-377 group law is mathematically proven on 3 independent points. The field arithmetic kernel passes 2,472 golden vectors. The Poseidon MDS matrix is machine-proven non-singular. The Sparse Merkle Tree is canonical-at-rest. The MSSC automaton exhibits deterministic finality and stall-breaking, now consuming the real threshold beacon. The BLS verification law $e(\sigma, G_2) = e(H, Y)$ is pairing-proven secret-free. The M2 order-bound core encrypts, orders under the real beacon, and decrypts by threshold shares. The sum-check engine, the commitment layer, and the NIFS fold exist in golden-proven Phase-0 form: two verified computations compress into one, exactly.

The architecture’s one-line soul: *ordering separated from knowledge (structural), safety separated from statistics (zeros, not probabilities), history separated from verification (one 73 KB proof), security priced in burned capital rather than probabilistic goodwill.*

A Decision Ledger Excerpt

The following table excerpts key decisions from the **DECISIONS.md** ledger (115 verified entries DEC-001 through DEC-115, plus citation-recovered decisions through DEC-180, and full rationale-bearing Sections VIII–X: DEC-181 through DEC-203):

ID	Component	Decision	Supersede
DEC-001	Arithmetic (P0)	Pasta curves, $L \approx 126$, unsigned mag + sign flags	N/A
DEC-002	Sequencing (M1)	Simulate-then-Commit via 192-node committee	N/A
DEC-003	Accumulator (P5)	Clamp-with-counter via $L' = 127$ scratchpad	N/A
DEC-004	Activations (P5)	Sigmoid domain ± 28 ; 1-LSB rule	N/A
DEC-005	Table Eval (P5)	Degree-2 chunk interpolation (T_0, T_1, T_2)	N/A
DEC-006	Lookup Auth (P5)	Standard LogUp pipeline	N/A
DEC-007	Rounding (P5)	RNTE strict witness constraints (q, r, b)	N/A
DEC-008	Integrity (P5)	arch_root, work_manifest, fidelity_root	N/A
DEC-012	Weights (S1)	Cap against uncapped total T_0 , single pass	S1-v1 §2.
DEC-013	Weights (S1)	Cluster-level W_{max} via payout-graph linkage	S1-v1 §2.
DEC-014	Weights (S1)	Φ eligibility: burned fees $\geq$ floor, non-cluster	S1-v1 §2.

DEC-015	Numerics (S1)	$W_i = \lfloor (\Phi^6 S^4)^{1/10} \rfloor$ via Newton root	S1-v1 §2.
DEC-016	Slashing (S1)	Canonical vote preimage <code>HSM_MSSC_VOTE_v1</code>	S1-v1 §4
DEC-017	Exit (S1)	Escrow $+E_{ev}$; checkpoint sync via P4 proofs	S1-v1 §5
DEC-019	Numerics (S1)	Unit normalization: $6a + 4b + \epsilon_g \leq 126$	S1-v2 §2.
DEC-025	Sequencing (M2)	Commit-then-Simulate (inverted pipeline)	M1-v1 co
DEC-028	Committee (M2)	$192+32$, $t = 112$; beacon sortition; ≤ 16 seats/cluster	N/A
DEC-030	Folding (P4)	Unified trust anchor: <code>decree_root</code> certified by BLS	N/A
DEC-031	Folding (P4)	Self-send routing constraint	N/A
DEC-032	Folding (P4)	Fee underflow protection (LT gate)	N/A
DEC-033	Folding (P4)	PAD routing bypass	N/A
DEC-034	Folding (P4)	Certificate assumption (BLS verified out-of-circuit)	N/A
DEC-039	Theory (A6)	Sequential quorum accumulation	N/A
DEC-181	Threshold (P-II)	Steps 7–10 decomposition; pairing deferred per DEC-032/034	N/A
DEC-189	Hash-to-G1	Counter preimage, even-y, cofactor-clear; $[r]H = \infty$	N/A
DEC-192	Beacon	<code>HSM_BEACON_V1</code> : 53-B preimage, threshold-signed chain	N/A
DEC-197	F_{q12} tower	$v^3 = \gamma = (5, 1)$, $w^2 = v$; tower-norm inverses	N/A
DEC-199	BLS law	$e(\sigma, G_2) = e(H, Y)$ secret-free; retires DEC-191 deferral	DEC-191
DEC-200	Retirement	<code>sim_beacon</code> retired; consensus goldens consume the real beacon	DEC-192
DEC-201	M2 KEM	G2-plane threshold KEM, unified $X_E = Y_2$; Phase-0 DEM	N/A
DEC-202	M2 records	$D_j = [S_j]R$ + order-bound attestations; pairing identity	N/A
DEC-203	M2 ordering	<code>ct_hash</code> , <code>sort_key</code> , <code>order_root</code> over the real beacon	N/A

B Glossary

CCS Customizable Constraint Systems — multivariate constraint framework generalizing R1CS/Plonkish/AIR

CycleFold Auxiliary circuit on the companion curve absorbing cross-curve commitments

DKG Distributed Key Generation — threshold key generation without trusted dealer

GKR Goldwasser-Kalai-Rothblum — interactive proof protocol for layered arithmetic circuits

HyperNova Multilinear sum-check-based folding protocol for CCS instances

IVC Incremental Verifiable Computation — recursive proof composition

LogUp Lookup argument via logarithmic derivative fractional identity

MSSC Metastable Sub-Sampled Consensus — Avalanche-family voting protocol

NIVC Non-uniform IVC — SuperNova-style folding across multiple circuit types

PCD Proof-Carrying Data — recursive proof chains

PoUW Proof of Useful Work — consensus weight from verified AI inference

RNTE Round-to-Nearest-Ties-to-Even — deterministic rounding mode

SMT Sparse Merkle Tree — authenticated data structure for keyed state